

INTRODUCING

CRIP

CLIMATE RISK INTELLIGENCE PLATFORM

Enter any address on Earth. Get a live risk score combining wildfire, earthquake, flood, weather, and air quality, with AI explaining what it means in plain language.

[Live Prototype link for CRIP](#)

DATA SOURCES

NASA · USGS · AQICN · Open-Meteo

AI ENGINE

Groq LLM

COST TO USER

Free. Always.



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DELHI RISK SCORE

HIGH

THE LANDSCAPE

Why climate risk intelligence must be democratized

THE CRISIS

Climate disasters are accelerating and the world is financially exposed

\$320B

Total economic losses from natural disasters in 2024. Only \$140B insured, \$180B borne by people and governments with no safety net.

Source: Munich Re NatCatSERVICE, Jan 2025

5x

 increase in disasters

Disasters increased five-fold (1970-2019). Losses rose 7x. 115 deaths and \$202M in losses every day for 50 years.

Source: WMO Atlas of Mortality (1970-2019)

91%

 of deaths in developing nations

The most affected are least equipped to prepare. Emerging Asia: 5% insured.

Source: WMO/UNDRR, Swiss Re Institute, 2026

THE MARKET

VCs and insurers betting billions on climate risk intelligence

\$1.8B → \$7.7B

Climate risk analytics at 17.6% CAGR through 2034. Broader market: \$7.4B → \$30.3B by 2035.

Source: Fortune Business Insights; Global Market Insights

WHO'S FUNDED

COMPANY	RAISED	PRICING
Jupiter Intelligence	~\$100M	\$100K-\$3M/yr
ClimateAI	\$38M	Enterprise only
Cervest	\$36.2M	Enterprise only
One Concern	\$22M+	Enterprise only
Moody's RMS	Acquired	Enterprise only

Investors: Sacca (Lowercarbon), Benioff, CDPQ, Liberty Mutual, QBE, Draper Esprit.

Sources: Crunchbase; Energize Ventures; TechCrunch

THE GAP

Every funded player serves Fortune 500s. Nobody built for the other 95%.

\$424B

Global protection gap in 2025. Grew from \$395B in one year — widening, not closing.

Source: Swiss Re Institute, 2025/2026

THE PRICING WALL

ENTERPRISE TOOLS TODAY

\$100K-\$3M/year · 6-month deployments · Fortune 500 only

CRIP

Free · 30 seconds · Anyone with a browser · Same data

UN VALIDATION

Sendai Target G: universal multi-hazard early warning by 2030. CRIP is Target G in product form.

Source: UNDRR Sendai Framework 2015-2030

FEMA/NIBS: Every \$1 in disaster mitigation saves \$6 in future costs.

Source: NIBS Mitigation Saves, 2017

PROBLEM FRAMING CANVAS

Five questions that validate why CRIP must exist



01 WHAT IS THE TRUE PROBLEM?

Climate risk data exists openly (NASA, USGS, NOAA) — but those who need it most cannot access or interpret it. An information asymmetry crisis.

6x higher disaster mortality without early warning systems
UNDRR MHEWS, 2025

83% of homebuyers seek climate risk data — Zillow removed scores under pressure
Zillow 2023; Scotsman Guide 2025

02 WHY SOLVE IT NOW?

Disasters up 5x, losses up 7x. The \$424B protection gap widens annually. UN Sendai deadline is 2030.

255 days of extreme weather in India in 2024 — 70% of the year
CSE, 2024

\$186B projected insured losses by 2030 at current growth rate
Swiss Re sigma 1/2025

03 HOW DOES IT GENERATE VALUE?

Every \$1 in mitigation saves \$6-\$13. CRIP converts the \$424B gap into free, actionable intelligence.

\$1 → \$6 FEMA-validated ROI on mitigation, up from \$4 in 2005
NIBS/FEMA, 2017

\$1.47T U.S. real estate at risk from climate change over 30 years
First Street 2025

04 WHO ARE THE CUSTOMERS?

120M+ Indian farming households, 83% of homebuyers, SMBs in 5% resilience markets, and enterprise API consumers.

120M+ Indian farming households hit by extreme weather. 41% PMFBY payout rate
Min. of Agriculture PIB 2024

5% insurance resilience in Emerging Asia — 95% of new assets uninsured
Swiss Re, 2026

05 HOW DO WE KNOW IT'S A PROBLEM?

\$320B losses, 83% demand, \$200M+ VC, UN/EU/SEC mandates, 6x mortality gap — five evidence layers.

\$200M+ VC into climate risk. Sacca, Benioff, CDPQ, Liberty Mutual
Crunchbase 2024

30% damage reduction from 24hr advance warning
UN SG 2025; UNESCAP

WHAT CRIP DOES ? AND WHY IT MATTERS



The same risk intelligence Fortune 500s pay \$100K/year for- accessible to anyone

BEFORE CRIP

How climate risk assessment works today

01 Locate the data yourself

Navigate 5+ government portals (NASA, USGS, NOAA, AQICN). Different formats, APIs, update frequencies.

02 Cross-reference manually

Download, clean, align coordinates, merge time-series. Requires GIS expertise most people don't have.

03 Hire a consultant

Environmental assessments: \$1,400-\$7,500/property. Enterprise platforms: \$100K-\$3M/year.

04 Wait weeks for results

Traditional assessments take 2-6 weeks. By delivery, the risk landscape has changed.



WITH CRIP

What the same process looks like now

01 Type any address

One input. Any location on Earth. No GIS knowledge. No account. Works on mobile and desktop.

02 5 APIs queried in real time

NASA FIRMS (fire), USGS (seismic), Open-Meteo (weather/flood), AQICN (air), Open-Elevation (terrain).

03 AI synthesizes plain-language report

Groq LLM processes all five data streams. Composite risk score + human-readable explanation. No jargon.

04 Results in 30 seconds. Free.

No consultant. No enterprise contract. Same government data the \$100K tools use, interpreted by AI.

WHY THIS IS DEFENSIBLE: THE DATA SOURCES ARE THE SAME

CRIP uses the exact same government data as enterprise platforms. The \$100K pays for packaging — not data.

NASA FIRMS

Wildfire

Active fire data within 3hrs of satellite overpass. MODIS + VIIRS. Free.

NASA Open Data Policy

USGS

Seismic

Global earthquake catalog. Real-time magnitude, depth, proximity. Public domain.

USGS Public Data

Open-Meteo

Weather & flood

Forecasts, historical data, river discharge. Open-source API. No key.

Open Source License

AQICN

Air quality

Real-time AQI from 1,000+ stations. EPA/WHO standard.

Open Data Project

Open-Elevation

Terrain

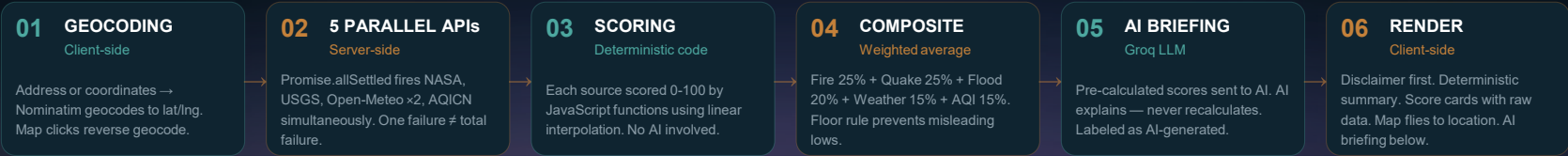
Global elevation for flood/landslide modeling. Public SRTM dataset.

Public Domain SRTM

HOW CRIP WORKS: ARCHITECTURE



6-step pipeline: address input → parallel data collection → deterministic scoring → AI interpretation → risk report



KEY ARCHITECTURAL DECISION

SCORES ARE CALCULATED BY CODE, NOT AI

The problem: Initial prototype had the AI model scoring risks. It gave Delhi 100/100 at midnight when most risks were low. LLMs hallucinate numbers.

The fix: All scoring moved to deterministic JavaScript functions using linear interpolation (lerp). Same data always produces the same score. The AI only explains scores it receives — it cannot modify them.

The result: Reproducible, testable, debuggable scores. Two explanation layers — one mathematically consistent (code), one natural language (AI, labeled as variable).

SCORING METHODOLOGY

Composite = weighted average of 5 risk dimensions

Wildfire

25%

Earthquake

25%

Flood

20%

Weather

15%

Air Quality

15%

Rationale: Fire & quake 25% each (immediately life-threatening), Flood 20% (property + life, localized), Weather & AQI 15% each (health, rarely fatal).

10 FAILURE MODES IDENTIFIED & ADDRESSED

MITIGATED	Missing hazard types	Prominent disclaimer listing coverage gaps	FIXED	No flood trend data	7-day discharge trend with directional arrows
MITIGATED	No empirical scoring basis	Scores labeled heuristic, methodology transparent	MITIGATED	Coordinate precision	Regional not property-level; coordinates shown
PARTIAL	No local baseline context	AI interprets locally; raw data shown alongside	FIXED	AI inconsistency	Deterministic summary as source of truth; AI labeled
FIXED	Different time windows	Each card shows its time window explicitly	FIXED	Data staleness	Scan timestamp on every result
MITIGATED	False sense of security	Disclaimer: silence ≠ absence of risk	FIXED	Scoring cliff edges	All functions use lerp — smooth curves, no cliffs

5 residual risks acknowledged — historical baselines, upstream freshness, flood proxy limitations, regional seismic norms, single-point assessment

PERFORMANCE & INFRA

< 5s

End-to-end response time including AI briefing

5

API calls fired in parallel — not sequentially

3x

Groq retry with exponential backoff (2s, 4s delays)

0

Infrastructure cost — Vercel serverless, auto-scaling

TECH STACK

React 18 · Vite · Tailwind · Leaflet · Vercel Serverless · Groq API

WHY IT MATTERS

The same risk intelligence that Fortune 500 companies pay \$100K/year for accessible to a farmer in Odisha, a homebuyer in California, a logistics company in Lagos.

"Substantially increase the availability of and access to multi-hazard early warning systems and disaster risk information to people by 2030."

— UN Sendai Framework, Target G

91% of disaster deaths occur in developing countries. Countries without early warning systems face 6x higher mortality. 24 hours of advance warning reduces damage by 30%.

Sources: WMO/UNDRR; UNESCAP; UNDRR MHEWS 2025

HOW TO MEASURE SUCCESS

ADOPTION

- Monthly active users
- Locations analyzed per day
- Geographic spread: are emerging markets using this?

ENGAGEMENT

- Return rate: do users scan before decisions?
- Session depth : do they read the AI briefing?
- Share rate- is this forwarded to others?

API TRACTION

- Developer signups
- API calls per month
- Revenue per call at scale

IMPACT PROXY

- % of scans from climate-vulnerable regions
- User surveys- did this change a decision?
- Pre-disaster vs post-disaster usage spikes

SCALING THE WEBAPP

NOW

Free web tool: portfolio + organic traction. Working prototype, global coverage, mobile responsive.

NEXT

API-as-a-service: developers pay per call. RESTful endpoint, JSON response, usage-based pricing like OpenWeatherMap.

THEN

Continuous monitoring + alerts; subscription model. Push notifications when risk changes for tracked locations.

LATER

Industry verticals :enterprise SaaS. Insurance portfolio scoring, real estate risk disclosure, agricultural early warning.

ADDITIONAL FEATURES ON ROADMAP

- Historical risk trends (10-year flood risk change)
- Multi-location portfolio scanning (10K+ properties)
- Climate-adjusted property valuation
- Embedded risk scoring for insurance/real estate platforms